

Initial research for GPS & Heart Rate data – Week 3

How are GPS & Heart Rate data used in football ?

GPS data allows to compute metrics that are needed to determine the **external load** (physical load) of a player, and **Heart Rate data** can be used to determine the **internal load** (psychological load) of a player in football. The computed metrics can then be used to assess a football player's workload management. In fact, excessive fatigue increases the risk of injury, especially when the external load exceeds the capacity of the athlete. For instance, in soccer (or football), **central fatigue** contributes directly to **anterior cruciate ligament (ACL) injuries**.

There are two methods to measure an athlete's internal load:

- **sRPE method (session-rating of Perceive Exertion)**: which requires the athlete to rate each session on a **10-point scale**, the load is then computed by multiplying the point-scale with the duration of the activity (**Load = RPE * duration in minutes**)
- **Heart Rate (HR) method**: which requires the use of a heart rate monitor, and is based on the **linear relationship** between the **HR** and the **rate of oxygen consumption (VO2)** during steady-state exercises. It also needs calibration of the athlete's HR training zones, and is pretty inaccurate when it comes to short, explosive activities.

Using activity trackers to measure and athlete's external load:

- **GPS/Accelerometers tracking devices**: they provide useful information regarding the athlete's work performed during the activity session. In fact, thanks to this **wearable technology**, it is possible to count the number of accelerations/decelerations, total covered distance, number of sprints, number of change in directions, etc...
- Here are some **metrics** that can help identify the **quality of workload management**:

While thresholds and indicators presented in this article are published in the scientific literature, they should be used as guideline and not as "magic" numbers. Large inter-individual variations exist.

Chronic Load (CL)

This is the average weekly load, typically calculated over the previous four weeks, but sometimes over a longer period. Usually, the higher the Chronic Load, the fitter and more robust the athlete is. In some situations, Chronic Load can be calculated using exponentially weighted moving averages.

Acute Load (AL)

The Acute Load represents the cumulative load of the current week. Usually, the higher the acute load (compared to chronic load), the more tired the athlete. In some situations, AL can also be calculated using shorter periods (e.g., three days).

Freshness Index (FI)

Similar to the Training stress balance proposed by Andrew Coogan¹⁷, the Freshness Index represents the difference between chronic and acute load (CL-AL) or between "fitness" and "fatigue." A positive Freshness Index indicates an unloading phase when low fatigue and good performance levels are to be expected.

Monotony

The Monotony Index proposed by Carl Foster³ measures the fluctuation of daily loads within the week. Intensive training combined with a high Monotony Index (>2) becomes an important risk factor for illness and overtraining³.

Strain

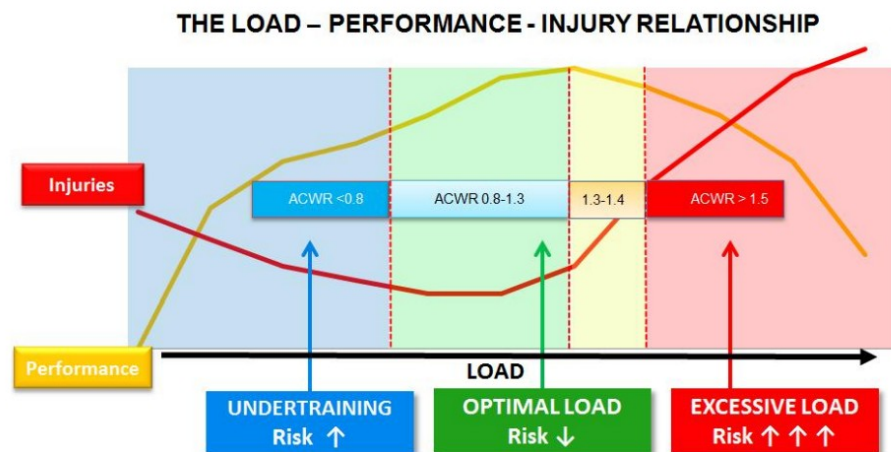
Early work by Foster demonstrated that 89% of illnesses and injuries could be explained by spikes in individual Strain in the 10 days preceding the incident³. Thus, monitoring the Strain variable can be a valuable technique for controlling individual adaptation to training load and preventing workload-related illness and overtraining³.

Acute:Chronic Workload Ratio

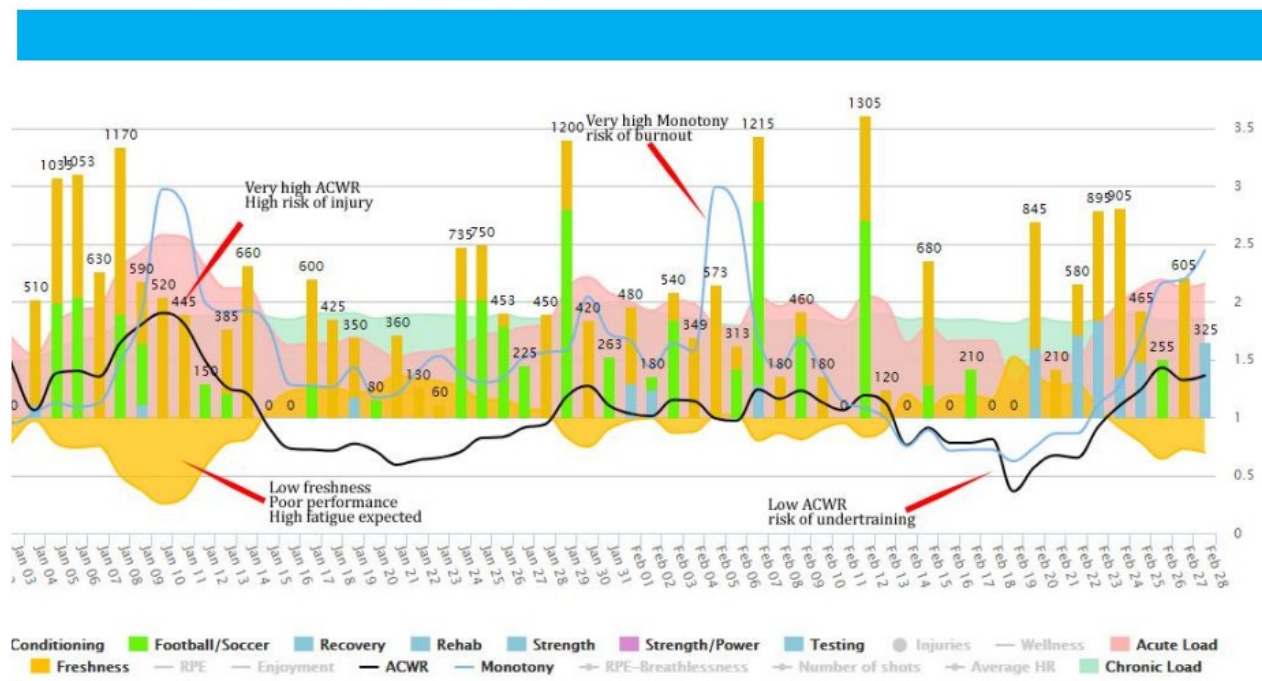
The Acute:Chronic Workload Ratio (ACWR)^{4,9,18} measures the relationship between acute load (current week load) and chronic load (average load of the last four weeks). The ACWR is a useful tool to detect spikes in load, for both internal and external load measures.

Monitoring ACWR helps to keep an athlete's workload in the high-load, low-risk zone (0.8-1.3). When ACWR is too low (less than 0.8) or too high (1.5 or more), risk increases and workload may need to be adjusted.

In order to find the **optimal workload management** for an athlete, the idea could be to **follow either of these analysis**:



Adapted from : Gabbett TJ., The training—injury prevention paradox: should players be training smarter and harder?, Br J Sports Med, 50:273–280, 2016
Figure 3 - Load-Performance-Injury Relationship



Type of data available in GPS/Heart Rate samples & definitions:

Here are the data types recorded by the trackers that will let us compute the **necessary parameters** in order to assess the **workload (internal & external loads)** of a football player, and construct the relevant graphs and representations to visualise at best the player's performance:

- **Time [in format HH/MM/SS.D]:** this value represents the exact timing during which the tracker recorded respective data relevant to the analysis
- **Latitude [in format GPS]:** this value represents a location's distance North/South from the Earth's equator, measured in degrees (0 → Equator, 90 → North/South poles)
- **Longitude [in format GPS]:** this value represents a location's distance East/West from the Prime Meridian, measured in degrees (0 → Prime Meridian, 90 → East/West poles, 180 → Anti-Meridian)
- **Speed [in m/s]:** this value represents the actual speed measured by the trackers

- **Heart Rate [in bpm]:** this value represents the actual beats per second measured by the sensor
- **Hacc (Horizontal Accuracy) [in unit]:** this value represents how confident is the GPS tracker regarding the reported GPS coordinate, and a smaller value indicates higher confidence as well as better accuracy
- **Hdop (Horizontal Dilution of Precision) [in unit]:** this value is calculated based on the positions of the GPS satellites currently visible to the receiver, and a smaller value (< 1) indicates better satellite geometry and thus better horizontal accuracy
- **Quality of Signal [in unit]:** this value represents how strong and accurate the received GPS signal actually is, and is affected by atmospheric conditions and multi-path interference
- **Number of Satellites [in units]:** this value represents the number of satellites visible to the GPS receiver at this point in time
- **Instantaneous Acceleration Impulse [in m/s^2]:** this value is computed by differentiating the instantaneous velocity measured by the GPS tracker using the Doppler shift
- **Accl X [in m/s^2]:** this value represents the player's acceleration measured in the X axis
- **Accl Y [in m/s^2]:** this value represents the player's acceleration measured in the Y axis
- **Accl Z [in m/s^2]:** this value represents the player's acceleration measured in the Z axis
- **Gyro X [in rad/s or degrees/s]:** this value represents the angular velocity measured by the Gyroscope around the X axis (roll)
- **Gyro Y [in rad/s or degrees/s]:** this value represents the angular velocity measured by the Gyroscope around the Y axis (pitch)
- **Gyro Z [in rad/s or degrees/s]:** this value represents the angular velocity measured by the Gyroscope around the Z axis (yaw)